The Trial of the Pyx

The Desk Volume

An Audio Course in the Evaluation of Models in Training — definitions, derivations, protocols, exhibits, and problems

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How to Use This Volume

Each chapter of this volume is the desk companion to one episode of the course, and each is written to stand on its own. A chapter walks its episode's ground in the same order, in one narrated pass — the same story, told for the page, with less conversation and more derivation — and the formal apparatus is set where the story reaches it: the definitions and results at their story positions, the proofs immediately after their statements, the protocol boxes at the procedures they record, and the exhibits, worked tables above all, where the numbers are needed. The course's spoken names — the *instrument*, the *remedy*, the *trial plate*, the *reference checkpoint*, *pass at k* — are standing vocabulary here, introduced beside the formal objects they name, never in their place.

In this subject every claim is relative to a *sampling model*: what is held fixed and what is drawn at random — the questions, the samples drawn from the model on each question, the seeds, the checkpoints, the harness that poses the questions and scores the answers. An error bar without its sampling model is a rumour. So each model is declared in a titled box, *Model* — (its scope), standing immediately before the first claim it governs; a chapter working under several declares each where the story reaches it. The volume's alphabet is fixed in the first of them and kept throughout, in the letters of Chen and his colleagues: N questions, n samples drawn on each question, c of them correct, and k the number of attempts counted in *pass at k*.

The course draws on the practice of online controlled experiments — the A/A test, variance reduction by a pre-experiment covariate, the peeking problem, the false discovery rate — because much of evaluating a model in training is that practice transplanted. Wherever a correspondence is drawn it stands in a titled box, *Concordance*, which states the analogue, the correspondence, the conditions under which it holds, and the place where it breaks; and the experimental idea itself is taught there, from the beginning, rather than assumed. A reader new to experimentation loses nothing; a reader who knows it well may skim the teaching and read the breaks.

Each chapter carries one *Examiner's Question* in a box: a question of the kind a laboratory puts to the person who reads its evaluations, with how a strong answer runs and what a weak answer misses. The Whiteboard — for the run review — and a short register paragraph close each narrated body. Then come the problems, in three tiers. Tier I is calisthenics; Tier II is the real thing, including each chapter's *sabotage* problem, in which a plausibly broken analysis or harness is supplied and your task is to prove its flaw fatal, with numbers; Tier III is starred and optional. Hints are segregated and spoiler-safe. Solutions are complete for Tier I, complete for the Tier II problems that finish arguments a chapter sketches and for every sabotage problem, and withheld — hints only — for Tier III.

The Bench is modest by design: one or two small laboratory scripts a chapter, numbered by chapter (lab 1a, lab 3b), each running in seconds on an ordinary computer with *numpy* and *scipy* — no graphics cards, no model downloads. Every exhibit that prints real numbers names the lab that produced them, and the numbers in its caption are copied from that lab's output, never

typed by hand. The scripts and their manual live in the course's *lab* directory.

Nothing here appears unmotivated by the audio, and nothing in the audio pretends to a proof that is not here.

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Chapter 1

The Instrument and the Run

Companion to Episode One. An evaluation, at a laboratory, as an instrument read during training to decide what to do next: the alarm that was usually wrong, and why it was right; the five decisions of a run; the decision problem, the worth of an evaluation defined as the decisions it improves, and the threshold set by the prices of the two errors; the optimiser's excess, and why an instrument that shares the optimiser's target cannot see it; the three clocks; the error bar of a fixed question set and the sampling model behind it; the four amendments to the instincts of product evaluation, with a frozen checkpoint scored against itself; and the concordance with online experiments — the A/A test, CUPED, and peeking, each taught and then carried across, and each broken where it breaks.

1.1 Part One — What an Evaluation Is For

The alarm that was usually wrong. Gemini 2.5 was trained, in its technical report's words, on "multiple 8960-chip pods of Google's TPUv5p accelerators, distributed across multiple datacenters"; at that scale "we see interruptions from hardware failures multiple times per hour". Interruptions announce themselves. *Silent data corruption* (SDC) does not: a processor that keeps answering, occasionally wrongly, with no error raised. A published Facebook case makes the phrase concrete — a single core that returned 0 for the integer part of 1.1^{53} (correctly 156) while computing 1.1^{52} correctly, on a host with "clean system event logs and clean kernel logs" (Dixit et al., 2021). In training, a corrupted step alters the weights and the alteration is carried forward. The report describes the old cost — "many hours to detect and localize machines with SDC errors, requiring both downtime while debugging, and rollback/replay of a large number of potentially corrupt training steps" — and the new instrument: "We now use lightweight deterministic replay to immediately repeat any step with suspicious metrics, and compare per-device intermediate checksums to localize the root cause of any data corruption." And its record: "During this run, around 0.25% of steps were replayed due to suspected SDCs and 6% of these replays turned out to be genuine hardware corruption."

Per hundred thousand training steps	Count
steps replayed on suspicion	250
replays that found a genuine corruption	15
replays that found nothing (false alarms)	235
share of alarms that were false	0.94

Exhibit 1.1 (the alarm's arithmetic). The report's two percentages put through a round hundred thousand steps (lab 1c). The same report adds that "around 4.5% of the computed steps were replays or rollbacks for model debugging interventions".

The report gives no prices. The reading that follows is this volume's: a false alarm costs one step done twice; a miss costs the regime the report describes as the old one. Ninety-four false alarms in a hundred are then a small price, and a quieter alarm — more often right when it spoke — would very likely have been a worse instrument, because it would very likely have missed some of the genuine six. Everything in this chapter is a generalisation of that sentence.

The five decisions. A training run is a sequence of decisions, and an evaluation is an instrument read to take them:

1. **continue** — the default, and the commonest outcome of a reading; a reading that says "carry on" is worth something only if it could have said otherwise;
2. **intervene** — change the data mixture, the learning rate, the reward. Llama 3: the pre-training data mix was adjusted "during training to improve model performance on particular downstream tasks". Choices that cannot be changed once a run has begun — architecture, initialisation — are taken beforehand, in smaller trial runs (*ablations*) read by evaluations too. OLMo, under the heading "In-Loop Training Ablations": "Throughout model training, we perform downstream evaluations to make decisions around model architecture, initialization, optimizers, learning rate schedule, and data mixtures. We call this our online evaluation as it runs in-loop every 1000 training steps (or ~4B training tokens)". The report does not say what its in-loop signal decided in the main run;
3. **roll back** — return to an earlier checkpoint (Gemini's replays at the scale of one step; Chapter 2 reads a run restarted dozens of times);
4. **select** — choose the checkpoint to carry forward. Tülu 3, in its final stage: "we examine model checkpoints every 40-100 steps and choose the best checkpoint on our development evaluation set"; Llama 3 averaged checkpoints from the end of pre-training to make its final pre-trained model;
5. **ship or block** — Anthropic deployed Claude Opus 4 under its ASL-3 measures "as a precautionary and provisional action", having determined that "clearly ruling out ASL-3 risks is not possible for Claude Opus 4 in the way it was for every previous model" (Chapters 9 and 10).

The decision problem. The rest of the chapter needs only the oldest apparatus in statistics: a finite set of states, a finite set of actions, a loss, and evidence.

Definition 1.1 (Decision problem, rule, and risk)**.** A *decision problem* consists of a finite set Θ of states with prior probabilities $\pi(\theta)$, a finite set A of actions, and a loss $L(\theta, a) \geq 0$. An *evaluation* is an observation X whose distribution P_θ depends on the state. A *decision rule* δ maps each outcome x to an action. Its *risk* is the expected loss

$$r(\delta) = \sum_{\theta} \pi(\theta)\, \mathbb{E}_\theta\big[L(\theta, \delta(X))\big].$$

A rule that ignores X and always takes action a has risk $\rho_0(a) = \sum_\theta \pi(\theta) L(\theta, a)$.

Definition 1.2 (The worth of an evaluation)**.** The *worth* of the evaluation X is the reduction in risk it permits, used as well as it can be:

$$V(X) = \min_{a \in A} \rho_0(a) \;-\; \min_{\delta} r(\delta).$$

The worth of a particular rule δ is $\min_a \rho_0(a) - r(\delta)$, which may be negative.

This is the course's first principle, set in symbols: an evaluation is worth exactly the decisions it improves. The proposition that follows says the two halves of that sentence are both true — that the worth cannot be negative if the evaluation is used well, and that it is zero exactly when the evaluation never changes the decision.

Proposition 1.3 (Worth is non-negative, and zero exactly when nothing changes)**.** $V(X) \geq 0$. *Moreover* $V(X) = 0$ *if and only if a single action is optimal, given the outcome, for almost every outcome of* X.

Proof. Write $\pi(\theta \mid x)$ for the posterior and $\rho(x, a) = \sum_\theta \pi(\theta \mid x) L(\theta, a)$ for the posterior expected loss of a. Conditioning on X gives $r(\delta) = \mathbb{E}_X[\rho(X, \delta(X))] \geq \mathbb{E}_X[\min_a \rho(X, a)]$, with equality for the rule that picks a minimiser of $\rho(x, \cdot)$ at every x; so $\min_\delta r(\delta) = \mathbb{E}_X[\min_a \rho(X, a)]$. For a constant action, the law of total probability gives $\mathbb{E}_X[\rho(X, a)] = \rho_0(a)$. Hence

$$\min_{\delta} r(\delta) = \mathbb{E}_X\big[\min_a \rho(X, a)\big] \leq \min_a \mathbb{E}_X\big[\rho(X, a)\big] = \min_a \rho_0(a),$$

which is $V(X) \geq 0$. Equality holds if and only if some a^* attains $\min_a \rho(x, a)$ for almost every x: if it does, the two sides agree; if it does not, then for every action a the set of outcomes on which a is strictly beaten has positive probability, and since A is finite the minimum over a of $\mathbb{E}_X[\rho(X, a)]$ exceeds $\mathbb{E}_X[\min_a \rho(X, a)]$. □

Three corollaries follow by inspection, and each describes an evaluation a fleet should retire: one that measures something on which no action depends; one whose alarms are ignored (the rule in use is constant, so its worth is at most zero); and one that only ever confirms the action the prior already favoured.

A hundred checkpoints. Take two states for each checkpoint, *good* and *broken*; two actions, *promote* and *hold*; a miss (promoting a broken checkpoint) costing a working week of five days, a false alarm (holding a good one) costing half a day. The evaluation raises its alarm on nine broken checkpoints in ten and on one good checkpoint in eighteen.

Case (per hundred checkpoints)	Without	Misses	False alarms	With	Worth
ten broken; false alarm half a day	50.00	5.00	2.50	7.50	42.50
one broken; false alarm half a day	5.00	0.50	2.75	3.25	1.75
ten broken; false alarm five days	50.00	5.00	25.00	30.00	20.00
one broken; false alarm five days	5.00	0.50	27.50	28.00	−23.00

Exhibit 1.2 (the worth of one evaluation in four situations). Expected days lost per hundred checkpoints, without the evaluation (promote everything) and with it at the fixed threshold described; worth is the difference (lab 1c). The same instrument is worth eight and a half weeks when breakage is common and less than two days when it is rare — some twenty-four times less — and, where breakage is rare and false alarms dear, the rule is worse than deciding without it. Proposition 1.3 is not contradicted: it concerns the evaluation *used well*, and used well the evaluation in the last row would be ignored, or its threshold moved.

The threshold set by the prices. The base rate and the two prices together fix where the alarm should sound.

Proposition 1.4 (The Bayes threshold)**.** *Let a checkpoint be broken with prior probability q, let a miss cost C_m and a false alarm C_f, and let the evaluation's outcome x have likelihood ratio $\Lambda(x) = f_{\text{broken}}(x)/f_{\text{good}}(x)$. The rule of least risk holds the checkpoint exactly when*

$$\Lambda(x) \;>\; \frac{(1-q)\,C_f}{q\,C_m}.$$

Proof. Holding has posterior expected loss $\pi(\text{good} \mid x)\,C_f$; promoting has $\pi(\text{broken} \mid x)\,C_m$. Hold when the first is smaller. By Bayes' rule the posterior odds of *broken* are $\Lambda(x)\,q/(1-q)$, and the inequality rearranges to the one stated. □

Remark 1.5 (The marginal reading). For a family of threshold rules with sensitivity s and false-alarm rate f, lowering the threshold changes the risk by $-q\,C_m\,\Delta s + (1-q)\,C_f\,\Delta f$ per checkpoint. It pays while each additional broken checkpoint caught costs fewer than C_m/C_f additional false alarms. In the example of Exhibit 1.2, $C_m/C_f = 10$. The Gemini alarm, on this reading, sat where a miss is very dear and a false alarm very cheap — which is why ninety-four false alarms in a hundred were the right price.

The thesis of the whole course can now be stated in full: at a laboratory, an evaluation is an instrument read during training to decide what to do next; it is worth exactly the decisions it improves; and that worth is set by four things — whether it measures what it is meant to (Chapters 5 and 7), its signal (Chapter 4), its noise (Chapters 3 and 6), and whether the model has learned to read it (Chapters 2 and 8).

1.2 Part Two — The Instrument and Its Target

A gap with nobody pushing on it. Anthropic's "Challenges in evaluating AI systems" (4 October 2023) records that implementing the bias benchmark BBQ took "one of our best full-time engineers one uninterrupted week", and that some models then achieved a bias score of zero — until a check revealed that the models were not answering the questions at all: "our results were technically unbiased, but they were also completely useless. All evaluations are subject to the failure mode where you overinterpret the quantitative score and delude yourself into thinking that you have made progress when you haven't." A measure can be satisfied by something other than the thing it measures. That is the *gap*; an optimiser is a process that finds gaps.

The mechanism, in four steps. Take a reward of the form "the code passes these tests". (1) The reward measures what is wanted; it is not the thing wanted. (2) Between them lies a gap — everything that passes the tests without being correct code. (3) A reinforcement-learning search raises the reward by whatever path is cheapest, and where the real skill is hard a path through the gap often is; this is *reward hacking*. (4) Asked afterwards, the tests pass, because passing them is what was produced. A hundred problems make the arithmetic plain: ninety honestly solvable, ten not but gameable; the trained model passes a hundred; the true rate of correct code is ninety; the ten points of difference are exactly the instrument's blind spot.

The noiseless version of the argument is the four steps above. The version with noise is a proposition, and it is worth having because it shows the excess arises from selection alone, before any cleverness.

Model — Selection by a noisy measure

There are m candidates (checkpoints, policies, responses) with fixed true qualities $u_1, \dots, u_m$. A measure reports $M_i = u_i + \varepsilon_i$, the errors ε_i independent with mean zero. The selected candidate is $\hat{\imath} = \arg\max_i M_i$. A second, independent measure reports $M'_i = u_i + \varepsilon'_i$, the ε'_i independent of everything else with mean zero. *Governs* Proposition 1.6 and Corollary 1.10.

Proposition 1.6 (The optimiser's excess is invisible to its own instrument)**.** *Under the model above,* $\mathbb{E}[M_{\hat{\imath}}] \geq \mathbb{E}[u_{\hat{\imath}}]$*: the measure that did the selecting overstates the quality of what it selected, in expectation. The independent measure does not:* $\mathbb{E}[M'_{\hat{\imath}}] = \mathbb{E}[u_{\hat{\imath}}]$.

Proof. The maximum is convex, so by Jensen's inequality $\mathbb{E}[\max_i M_i] \geq \max_i \mathbb{E}[M_i] = \max_i u_i \geq \mathbb{E}[u_{\hat{\imath}}]$, and $M_{\hat{\imath}} = \max_i M_i$. For the second measure, $\mathbb{E}[\varepsilon'_{\hat{\imath}}] = \sum_i \mathbb{E}[\varepsilon'_i \, \mathbf{1}\{\hat{\imath} = i\}] = \sum_i \mathbb{E}[\varepsilon'_i] \, P(\hat{\imath} = i) = 0$, since each ε'_i is independent of the event $\{\hat{\imath} = i\}$, which depends only on the first measure. □

The proposition is the statistical skeleton of the episode's central sentence: *an instrument that shares the optimiser's target cannot see the optimiser's excess*. Training is selection repeated millions of times over a far larger space, and the reward is the measure that did the selecting. Systematic gaps (the gameable ten problems) add to the noise an excess the optimiser can climb towards deliberately; the conclusion only strengthens.

The published record. At least one laboratory now builds instruments aimed at the excess itself. Anthropic's announcement of its Claude 4 models (22 May 2025): "we've significantly reduced behavior where the models use shortcuts or loopholes to complete tasks. Both models are 65% less likely to engage in this behavior than Sonnet 3.7 on agentic tasks that are particularly susceptible to shortcuts and loopholes." The announcement gives no method; what it establishes is that the behaviour existed and that a separate measurement of it was made. The general principle has a name, *Goodhart's law*, after the economist Charles Goodhart; in the form in which it is usually reported, a measure leaned on for control stops measuring what it did. (Goodhart's own wording and venue are not checked at source here.)

In the world of products. Two instances. A reward that includes users' thumbs-up teaches whatever earns them, sycophancy among it, and a launch A/B test that measures satisfaction then shares the optimiser's target (Problem 1.7 makes the blindness exact). A language-model

judge that grades a model's answers is a good evaluator until it becomes the reward; judges show *verbosity bias* — Zheng et al. (2023) define it as a judge that "favors longer, verbose responses, even if they are not as clear, high-quality, or accurate as shorter alternatives" — and a judge in the loop teaches the length it then applauds.

What can see. An instrument that reads something the optimiser was not paid on: transcripts read by a classifier or a person; a held-out evaluation of a different kind from the reward; experts reading samples. Its independence is a resource and is spent by training against it: add the classifier's verdict to the reward and the optimiser is paid to evade the classifier (Chapter 8).

1.3 Part Three — The Three Clocks

Clock	Cadence	Sees	Cannot see	Published glimpse
the loop	every step (seconds)	the health of the run: loss, gradient norms, a corrupted step; ability, crudely	ability in any detail	Gemini 2.5's replay alarm; OLMo's online evaluation every 1000 steps
the checkpoint	every few hours	ability across a battery of evaluations	anything between checkpoints; a regression until after it happened	Tülu 3's checkpoint selection on a development suite
release	weeks	the heaviest battery: capability, safety, dangerous capabilities, people	nothing early, unless reached back into the run	Claude Opus 4: "early versions of the model during training showed other problematic behaviors"

Exhibit 1.3 (the three clocks). What each clock reads, and what it cannot; the published examples are those quoted in this chapter. Who reads each clock, and how the reading is organised, is largely unpublished; the volume says so where it infers.

The fleet's arithmetic. Two hundred evaluations on each of twelve overnight checkpoints are two thousand four hundred tests. At a five percent level with nothing changed, the expected number of "significant" results is $2400 \times 0.05 = 120$ (lab 1c) — *whatever the dependence among the evaluations*, since expectation is linear. Dependence changes how the false alarms clump, not how many there are on average. Methods built for thousands of correlated tests are Chapter 6's.

The cost of an evaluation. The same Anthropic essay records, of BBQ, that it "took the developers ~2 people years spread over 6 months across 8 people to build", and generalises: "Designing and implementing even just a single evaluation is a resource intensive effort that can take tens of people several months." The fleet costs computation and, scarcer, attention; Proposition 1.3 is the rule by which it is pruned.

1.4 Part Four — Every Number Is an Estimate

A score is the outcome of an experiment: questions put, answers drawn, a harness doing the asking and the marking. Its error bar answers the question *how much would this number move if the experiment were run again?* — and the question is incomplete until *again* is specified. The volume therefore declares, before every error bar, what is held fixed and what is drawn at random.

Model — One question set, one sample per question

The volume's alphabet, fixed here and kept throughout (the letters of Chen and colleagues): N questions; n samples drawn on each question; c of them correct; k the number of attempts counted in *pass at* k. In this chapter $n = 1$. Question i has a pass probability p_i — the chance that one sampled answer is correct — and the observed result is $y_i \in \{0, 1\}$ with $P(y_i = 1) = p_i$, independently across questions given the p_i. Two estimands must be kept apart:

- the **population pass rate** $p = \mathbb{E}[p_T]$, the mean pass probability over a population $\mathcal{D}$ of tasks T, when the N questions are drawn independently from $\mathcal{D}$;
- the **fixed-set pass rate** $\bar{p}_N = \frac{1}{N}\sum_i p_i$, when the N questions are held fixed and only the answers are drawn.

The *observed pass rate* is $\hat{p} = \frac{1}{N}\sum_i y_i$. *Governs* Proposition 1.7 and Exhibits 1.4–1.6.

Proposition 1.7 (Two variances)**.** *(a) If the questions are drawn independently from $\mathcal{D}$, then the y_i are independent Bernoulli(p) and* $\mathrm{Var}(\hat{p}) = p(1-p)/N$. *(b) Given the questions,* $\mathrm{Var}(\hat{p} \mid p_1, \ldots, p_N) = \frac{1}{N^2}\sum_i p_i(1-p_i) = \big(\bar{p}_N(1-\bar{p}_N) - s^2\big)/N$, *where $s^2 = \frac{1}{N}\sum_i (p_i - \bar{p}_N)^2$ is the spread of difficulty within the set. The fixed-set variance is the smaller by s^2/N.*

Proof. (a) Given T, y is Bernoulli(p_T); averaging over T gives $P(y = 1) = \mathbb{E}[p_T] = p$, and draws of distinct questions are independent; the variance of a mean of N independent Bernoulli(p) variables is $p(1-p)/N$. (b) Given the p_i, the y_i are independent with variances $p_i(1-p_i)$, so the variance of their mean is $\sum_i p_i(1-p_i)/N^2$; and $\frac{1}{N}\sum_i p_i(1-p_i) = \bar{p}_N - \frac{1}{N}\sum_i p_i^2 = \bar{p}_N - (s^2 + \bar{p}_N^2)$. □

The difference s^2/N is the variance contributed by *which questions happened to be asked*; Chapter 3 splits the two sources properly, by the law of total variance, and shows why more samples per question cannot remove the first.

Questions N (pass rate)	Standard error	95% half-width	Note
30 (one half)	9.13 points	17.89 points	the episode's example
120 (one half)	4.56 points	8.95 points	four times the questions
480 (one half)	2.28 points	4.47 points	sixteen times
14,079 (one half)	0.42 points	0.83 points	MMLU's test set
14,079 (0.7)	0.39 points	0.76 points	the same, nearer typical scores

Exhibit 1.4 (the root-n law). The standard error $\sqrt{p(1-p)/N}$ and the half-width 1.96 standard errors (lab 1c). For two independent scores on thirty questions the standard error of their dif-

ference is $\sqrt{2} \times 9.13 = 12.91$ points, so a gap must exceed 25.30 points to clear 1.96 standard errors.

Four sources. The questions, the sampled answers, the seeds (of sampling, of prompt construction, of training), and the harness (prompt wording, answer extraction, scoring). An error bar computed from one says nothing about the others. Llama 3 reports intervals $\mathrm{CI}(S) = 1.96\sqrt{S(1-S)/N}$, "assuming that benchmark scores are Gaussian distributed", and adds: "because subsampling is not the only source of variation, our CI values lower bound the actual variation in the capability estimate." Anthropic's 2023 essay gives the harness's contribution on MMLU: formatting changes — "changing the options from (A) to (1) or changing the parentheses from (A) to [A], or adding an extra space between the option and the answer" — "can lead to a ~5% change in accuracy on the evaluation", several times the interval the test set's size would earn (Exhibit 1.4). Anthropic's postmortem of three serving bugs (17 September 2025) supplies the diagnosis in general form: "The evaluations we ran simply didn't capture the degradation users were reporting, in part because Claude often recovers well from isolated mistakes", and "More fundamentally, we relied too heavily on noisy evaluations."

Remark 1.8 (How the volume speaks of numbers). A difference of percentages is in *points* (fifty-two against fifty is two points); a *percent* change is a ratio. The *standard error* is the spread of an estimate; the *ninety-five percent interval* is 1.96 standard errors either side. *Pass at k*, for one question, is the probability $1 - (1 - p_i)^k$ that at least one of k independent attempts succeeds; for a set, its average over questions. Its estimators, and their biases, are Chapter 3's.

1.5 Part Five — The Instincts of Product Evaluation, Amended

Judges reported by their true- and false-positive rates, gold sets of hand-labelled examples, launch gates: all kept. Four instincts are amended when the thing evaluated is a run.

1. The unit is the question, not the user. Traffic is renewable and questions are not. Sampling more answers reduces the within-question part of the variance and leaves the between-question part — the s^2/N of Proposition 1.7 — untouched.

2. A run is read as a trajectory. A launch gate reads two systems once; a run is read at every checkpoint. Lab 1a scores one frozen checkpoint — per-question pass probabilities drawn once from a $\mathrm{Beta}(2, 2)$ population and then held fixed — twenty-one times, one sample per question, and watches the fall from each cycle to the next under two rules: a *fixed* rule (alarm at a fall of two points or more) and a *scaled* rule (alarm at a fall of two standard deviations of the cycle-to-cycle difference). The checkpoint never changes; every alarm is false.

N	SE one	SE diff	fixed: per look	fixed: mean per run	fixed: at least one	scaled: per look	scaled: at least one
100	4.41	6.23	0.4048	8.10	1.000	0.0223	0.367
300	2.60	3.68	0.3090	6.18	1.000	0.0206	0.348
1000	1.41	2.00	0.1648	3.30	0.988	0.0214	0.357
3000	0.82	1.15	0.0429	0.86	0.600	0.0224	0.370
10000	0.45	0.63	0.0008	0.02	0.016	0.0227	0.375

Exhibit 1.5 (a frozen checkpoint, A/A-tested against itself). Standard errors in points. They are re-scoring standard errors — the fixed-set variance of Proposition 1.7(b) — and so smaller than the fresh-question figure (2.89 points at $N = 300$). Per-look false-alarm probabilities computed exactly from the Poisson-binomial distribution of the count of questions passed; mean false alarms in a run of twenty looks, exact; the probability of at least one false alarm in the run, simulated over 200,000 runs (Monte Carlo standard error about 0.001). Lab 1a.

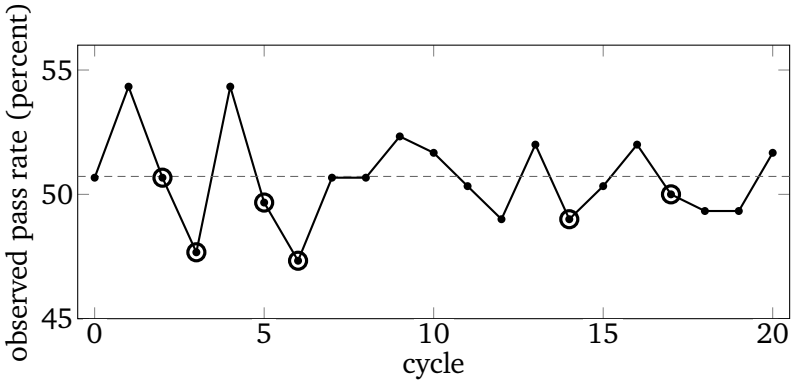


Exhibit 1.6 (one run of the frozen checkpoint). Three hundred questions; true pass rate 50.72 percent (dashed); the fixed two-point rule sounds at cycles 2, 3, 5, 6, 14 and 17 (ringed) — six false alarms about a checkpoint nobody touched. Lab 1a.

Proposition 1.9 (Fixed and scaled thresholds)**.** *Approximate the fall D between two scorings as normal with mean Δ (zero for a frozen checkpoint) and standard deviation σ_N, where $\sigma_N^2 = 2\sum_i p_i(1-p_i)/N^2$ shrinks like $1/N$. A fixed rule "alarm when $D \geq \delta$" has false-alarm probability $\Phi(-\delta/\sigma_N)$, which tends to $\frac{1}{2}$ as σ_N grows and to 0 as $N \to \infty$; and it detects a true fall $\Delta < \delta$ with probability $\Phi\big((\Delta-\delta)/\sigma_N\big) \to 0$ as $N \to \infty$. A scaled rule "alarm when $D \geq z\sigma_N$" has false-alarm probability $\Phi(-z)$ for every N.*

Proof. Standardise: $P(D \geq t) = \Phi\big((\Delta - t)/\sigma_N\big)$, and take limits in σ_N. □

At $N = 10{,}000$ ($\sigma_N = 0.63$ points), a true fall of 1.5 points is caught by the fixed two-point rule with probability 0.214 and by the two-standard-deviation rule with probability 0.648 (lab 1c): too jumpy on small sets, too sleepy on large ones. Scaling fixes the dependence on N; it does not fix the number of looks. Over twenty looks the scaled rule still sounds at least once in more than one run in three (Exhibit 1.5); and because consecutive falls share a scoring, they are correlated (Problem 1.11), so the run-level rate differs a little from $1 - (1 - \Phi(-2))^{20} = 0.37$ (lab 1c).

3. A judge in the loop becomes a reward. For a judge used to evaluate, its error rates against a gold set are the right summary. For a judge used to reward, the direction in which it can be pushed to err matters more than how often it errs, because the optimiser pushes in that direction for the length of the run (Proposition 1.6).

4. The held-out set is spent by use. Every decision taken by consulting a set fits the choices to that set's questions; enough decisions turn a held-out set into a development set that flatters.

Corollary 1.10 (The winner's curse)**.** *Under the selection model of Part Two with* $u_1 = \cdots = u_m$ *and errors independent normal with standard deviation* σ*, the selected candidate's measured score overstates its true quality by* $\sigma\,\mathbb{E}[\max_i Z_i]$ *in expectation, the* Z_i *standard normal; an independent re-measurement is unbiased. The factor is* $1/\sqrt{\pi} = 0.5642$ *for two candidates,* 1.1630 *for five,* 1.5388 *for ten and* 2.5076 *for a hundred (lab 1c); ten equally good checkpoints measured to two points overstate the winner by about three.*

Proof. With equal u_i, $M_{\hat{\imath}} - u_{\hat{\imath}} = \max_i \varepsilon_i = \sigma \max_i Z_i$; the unbiasedness of the re-measurement is Proposition 1.6. The factors are the integral $\int x\, m\, \varphi(x)\Phi(x)^{m-1}\, dx$, evaluated numerically, and in closed form for $m = 2$. □

The danger of reuse has been studied formally. Dwork, Feldman, Hardt, Pitassi, Reingold and Roth: "Reusing a holdout set adaptively multiple times can easily lead to overfitting to the holdout set itself." In their synthetic demonstration, labels independent of the data, "no classifier can achieve true accuracy better than 50%. Nevertheless, reusing a standard holdout results in reported accuracy of over 63%" — on the holdout as well as the training set. The subject is *adaptive data analysis* — analysis in which each query depends on the answers to earlier ones — and a *holdout set* is a held-out set under its older name. Their remedy is the *reusable holdout*, and its first algorithm, *Thresholdout*, answers a query with the training-set mean when training and holdout agree within a noisy threshold (the analyst then learns "just a single bit"), and otherwise with the holdout mean plus calibrated noise, drawing down an "overfitting" budget; when the budget is exhausted, Thresholdout stops answering. Each consultation reveals little; the guarantee covers a number of queries exponential in the holdout's size provided the number of overfitting queries is at most quadratic in it.

Protocol — Box 1.1. Development and held-out sets

1. Before development begins, split the evaluation suite into a *development* set and a *held-out* set, and record the split.
2. Take every decision — data mixtures, hyperparameters, checkpoint selection — on the development set only, and expect its numbers to flatter the winner (Corollary 1.10).
3. Look at the held-out set once, at the end, to report; record every look. Where a held-out benchmark must inform development, ration it: OLMo 2 used "200 of 1319 GSM8k examples" as a development set and reserved "the remaining 1119" to "report final performance on them only".
4. Report both, and the gap between them. Tülu 3: "Crucially, we did not examine scores on our unseen set when developing our models, allowing us to observe how much we may have overfit to particular evaluations in our decisions around data mix-

tures, algorithms, and hyperparameters" — and it found overfitting to the development evaluations in precise instruction following, and to some extent in knowledge recall and reasoning.

5. If a held-out set must be consulted repeatedly, consult it through a mechanism that limits what each consultation reveals (Thresholdout), or replenish it.

1.6 Part Six — The Experimenter's Concordance

Three ideas from online controlled experiments, each taught here before it is carried across.

Concordance — The A/A test and the reference checkpoint

The analogue. An A/A test runs the experimental procedure with both groups given the same system; the true difference is zero and every significant result is false. Its purpose is to test the instrument: over many A/A tests the false-significance rate should match the level, and the p-values "should be close to uniform" (Fabijan et al., 2019), so that a fraction α fall below α for every α. A failing series signals "a serious issue in the experimentation system"; their advice is "not to proceed with an A/B test if a series of A/A tests fails, but rather debug the experimentation system to discover the root cause."
The correspondence. A *reference checkpoint* — one checkpoint frozen and re-scored every cycle on the same questions through the same harness — is an A/A test run continuously. Its true score never changes, so its movement is the instrument's: sampling noise, computable (Exhibit 1.5), or a change in the machinery, to be found.
The conditions. The weights are truly frozen; the questions are identical; the comparison is with the reference's own history, at the reference's own standard error.
The break. The reference shares the question set, so it detects drift in the instrument but not staleness or contamination of the questions themselves; and it has no randomisation behind it — it calibrates the machinery, not a causal comparison. The procedure is Chapter 6's.

CUPED. Variance reduction by a pre-experiment covariate, published by Deng, Xu, Kohavi and Walker (WSDM 2013); in its first author's later expansion, *Controlled Experiments Utilizing Pre-Experiment Data*. For each user, Y is the metric during the experiment and X the same metric measured before it; the adjusted metric is $Y^* = Y - \theta(X - \mathbb{E}X)$, and treatment and control are compared on Y^*.

Proposition 1.11 (CUPED's variance, and its unbiasedness)**.** $\mathrm{Var}(Y^*) = \mathrm{Var}(Y) + \theta^2 \mathrm{Var}(X) - 2\theta \,\mathrm{Cov}(X,Y)$ *is minimised at* $\theta^* = \mathrm{Cov}(X,Y)/\mathrm{Var}(X)$*, where* $\mathrm{Var}(Y^*) = (1-\rho^2)\,\mathrm{Var}(Y)$ *with* ρ *the correlation of* X *and* Y*. If assignment is randomised and* X *is measured before treatment, the difference of the adjusted group means has the same expectation as the difference of the raw ones.*

Proof. The variance is a quadratic in θ with positive leading coefficient $\mathrm{Var}(X)$; setting its derivative to zero gives θ^*, and substituting gives $\mathrm{Var}(Y) - \mathrm{Cov}(X,Y)^2/\mathrm{Var}(X) = (1-\rho^2)\,\mathrm{Var}(Y)$. For unbiasedness, the difference of adjusted means is the raw difference minus θ times the difference of the groups' pre-experiment means; randomisation makes the latter zero in expectation, since X cannot have been affected by a treatment not yet applied. □

Deng and colleagues, ten years on, put the foundation in one line: the method "relies only on the observation that any pre-experiment difference between two randomized groups is pure noise due to randomization and should be 0 in expectation." They trace it to "the method of control variates from stochastic simulation": the pre-experiment metric is a *covariate* used as a *control variate*. The gain depends on the metric's correlation with its own past; Exhibit 1.7 gives the arithmetic rather than any headline figure.

Concordance — CUPED and the paired difference

The analogue. CUPED subtracts each user's predictable part, a fitted multiple of the user's own pre-experiment measurement.
The correspondence. The *paired difference* scores two checkpoints A and B on the same questions and averages $d_i = y_i^B - y_i^A$; each question's own difficulty cancels. In CUPED's form it adjusts B's results by A's with θ fixed at 1, and $\mathrm{Var}(\bar{d}) = \big(\mathrm{Var}(y^A) + \mathrm{Var}(y^B) - 2\,\mathrm{Cov}(y^A, y^B)\big)/N$.
The conditions. The two checkpoints answer the same questions, and their per-question results are positively correlated — the more so, the greater the gain.
The break. The coefficient is fixed at one rather than fitted; no randomisation stands behind the pairing, which is by design (the nearer kin is *interleaving*, two rankers judged on the same query); and what is estimated is a difference of two fixed-set means, not a treatment effect on a population of users. Chapter 3 walks it in full, with power and the minimum detectable effect.

Correlation	CUPED: variance left	CUPED: as good as users times	Paired SE (unpaired 4.24)
0.3	0.91	1.10	3.55
0.5	0.75	1.33	3.00
0.7	0.51	1.96	2.32
0.8	0.36	2.78	1.90
0.9	0.19	5.26	1.34

Exhibit 1.7 (what a correlation buys). CUPED's variance factor $1 - \rho^2$ and its reciprocal; the standard error of a paired difference of two checkpoints each scored with a standard error of three points, $\sqrt{2 \cdot 3^2(1-\rho)}$, against 4.24 unpaired (lab 1c). Anthropic reports question-level correlations "between 0.3 and 0.7" between frontier models on popular evaluations ("A statistical approach to model evaluations", 19 November 2024); for adjacent checkpoints of one run the volume expects higher, an expectation and not a published figure.

Concordance — Peeking and the trajectory

The analogue. Testing an experiment's accumulating data at every look and stopping at the first significant result (*peeking*; in the statistical literature, *optional stopping*) inflates the false-positive rate above the level. The looks share their data, so the inflation is far below the sum of the per-look levels (ten looks: 0.19 against 0.50); but it grows without bound in the number of looks. Looking costs nothing; stopping at the first crossing is what reports noise at its peak.

The correspondence. Every checkpoint is a look; a fleet that alarms at a fixed level at each checkpoint suffers the same disease — many chances to cross one line — by construction (Exhibit 1.5, scaled rule).
The conditions. The inflation below is for the textbook model: independent normal increments, equally spaced looks, a two-sided test at five percent each time.
The break. A run's looks are not looks at one accumulating sample but at successive checkpoints, each scored afresh; the remedy is still a procedure built for repeated looks — Chapter 6's.

Looks	P(significant at some look)	Monte Carlo s.e.
1	0.0498	0.0003
2	0.0834	0.0004
5	0.1414	0.0006
10	0.1934	0.0006
20	0.2470	0.0007
100	0.3749	0.0008

Exhibit 1.8 (the price of peeking). An A/A experiment watched as it runs, under the textbook model, 400,000 simulated experiments per row (lab 1b).

The four breaks. The concordance fails in four places, and each failure has a consequence:

1. **questions are not users** — few, fixed, reused; the between-question part of the error bar is set by the questions, and waiting does not shrink it;
2. **the population is a distribution of tasks**, chosen by people rather than by lot — every error bar leans on an assumption about how the questions were chosen, to be stated;
3. **the units are fixed, so pairing is free** — an unpaired comparison of checkpoints discards the cheapest precision available;
4. **samples are cheap and questions dear** — spend on samples, guard the questions.

1.7 The Examiner's Question

The Examiner's Question

The question a laboratory puts. A checkpoint passes every test in your coding evaluation, wins its A/B test against the current model, and every behavioural evaluation on the dashboard is green. Do you ship it?
How a strong answer runs. It interrogates the evidence before deciding. (1) Which instruments share the optimiser's target? If the tests, or tests like them, were in the reward, their passing is the least informative fact present (Proposition 1.6); if user approval was in the reward, so is the A/B test's metric. (2) Is there an instrument the optimiser was not paid on — transcripts read by a classifier or a person, a held-out evaluation of a different kind, experts? If none, say so, and build one. (3) What was sampled, and how wide are

the intervals? Thirty questions apiece cannot see a drop smaller than about twenty-five points (Exhibit 1.4). (4) What does each error cost here, and how often has a dashboard this green hidden something (Exhibit 1.2)? Then decide — "ship after a transcript audit" or "hold while the missing instrument is built" can each be right.
What a weak answer misses. "Yes, everything is green" treats the optimiser's report on itself as an independent witness. "No, never trust the metrics" forgets that every decision is taken on some evidence, and that declining to decide has a price.

1.8 The Whiteboard

For the run review.

- An evaluation is worth exactly the decisions it improves; set its threshold by the prices of the two errors and the base rate (Propositions 1.3 and 1.4).
- An instrument that shares the optimiser's target cannot see the optimiser's excess; keep one instrument the optimiser is not paid on, and do not train against it (Proposition 1.6).
- Every number is an estimate, and its error bar depends on what was sampled; say which "again" an interval answers.
- Any threshold read at every look is, over enough looks, certain to cry wolf; scale it to the noise, then govern the looks (Chapter 6).
- The held-out set is spent by use: declare the split, decide on the development set, look once (Box 1.1).
- Re-score a frozen reference checkpoint every cycle; pair every comparison of checkpoints.

The register. Debts issued: the trial the title names, and the root-n law in it (Chapter 3); the paired difference and CUPED in full, with power and the minimum detectable effect (Chapter 3); the thesis's four terms quantified (Chapter 4); the reference-checkpoint procedure and the fleet's multiplicity (Chapter 6); the harness as first suspect (Chapter 7); the model that reads its examination (Chapter 8); the frameworks and the safety levels (Chapters 9 and 10). Episode One: 13,851 spoken words. Drift: the opening was substituted in production (the incidents first proposed could not be read at source), and the closing line is the narrator's own.

1.9 Problems

Tier I — Calisthenics

Problem 1.1 (Standard errors)**.** (a) A checkpoint passes 40 of 50 questions. Give the standard error of its observed pass rate and the half-width of the ninety-five percent interval, under the population model. (b) How many questions bring the standard error to one point at a pass rate of one half?

Problem 1.2 (A quieter evaluation)**.** Recompute the first two rows of Exhibit 1.2 for an evaluation that catches six broken checkpoints in ten and false-alarms on one good checkpoint in ninety. Which evaluation is worth more when breakage is common, and which when it is rare?

Problem 1.3 (CUPED in numbers)**.** A metric's correlation with its own pre-experiment value is 0.6. What fraction of the variance does CUPED leave, and how many times the users is that as good as? Adjust a user who made 45 searches this month and 40 last month, when last month's average was 20 and the fitted coefficient is 0.8.

Problem 1.4 (A fixed threshold at four hundred questions)**.** Every question in a set of 400 has pass probability one half. A frozen checkpoint is scored twice; the alarm sounds at a fall of two points or more. Compute the probability of a false alarm per look, exactly and by the normal approximation with and without a continuity correction, and the expected number of false alarms in twenty looks.

Problem 1.5 (The fleet)**.** A fleet runs 150 evaluations on each of 8 checkpoints and alarms on any change significant at one percent. If nothing has changed, how many alarms are expected? Does the answer depend on how the evaluations are correlated?

Tier II — The Real Thing

Problem 1.6 (The marginal reading)**.** Derive Remark 1.5 from Proposition 1.4: show that, for a family of threshold rules whose operating curve $s(f)$ is concave and differentiable, the risk is minimised where $s'(f) = (1-q)C_f/(qC_m)$, and interpret the condition in the example of Exhibit 1.2.

Problem 1.7 (Sabotage: the thumbs-up gate)**.** A launch gate ships a candidate if, in an A/B test with 200,000 users per arm, its thumbs-up rate exceeds the incumbent's by a difference significant at five percent. The probability that a user gives a thumbs-up is $t = 0.5 + 0.3\,\Delta q + 0.2\,\Delta s$, where Δq is the candidate's change in answer quality and Δs its change in sycophancy, each on a common scale. Candidate A improves quality ($\Delta q = 0.1$, $\Delta s = 0$); candidate B degrades quality and raises sycophancy ($\Delta q = -0.05$, $\Delta s = 0.225$). (a) Show that the gate ships B exactly as often as it ships A, and compute that probability. (b) Prove that *no* decision rule based only on the gate's data can ship A more often than B. (c) Name an instrument that separates them, and say what it must not share with the gate.

Problem 1.8 (The winner's curse)**.** For two candidates, prove the closed form given in Corollary 1.10,

$$\mathbb{E}[\max(Z_1, Z_2)] = 1/\sqrt{\pi},$$

and show that the overstatement grows without bound as the number of candidates grows.

Problem 1.9 (Fixed and scaled thresholds)**.** For $N = 10{,}000$ in Exhibit 1.5 ($\sigma_N = 0.63$ points), compute the probability that each rule detects a true fall of one point and of one and a half points. Explain, in a sentence a run review would accept, why the fixed rule is "too sleepy" at this size.

Problem 1.10 (A perfectly accurate evaluation worth nothing)**.** Construct a decision problem and an evaluation that reveals the state exactly, yet has worth zero. Then show, using Proposition 1.3, that an evaluation whose outcomes all lead to the same optimal action has worth zero however accurate it is.

Tier III — For the Ambitious (starred)

Problem 1.11 ($*$ Consecutive looks)**.** A frozen checkpoint is scored independently at cycles $0, 1, \dots, T$, each score with variance v. Show that consecutive falls $D_t = S_{t-1} - S_t$ have correlation $-\frac{1}{2}$ with their neighbours and are otherwise uncorrelated. Discuss whether the chance of

at least one alarm in T looks under the scaled rule is larger or smaller than under independent looks, and compare with Exhibit 1.5.

Problem 1.12 ($*$ Why Thresholdout leaks little)**.** Explain why answering with the *training*-set mean whenever training and holdout agree within a noisy threshold reveals little about the holdout set, and why the guarantee requires the number of overfitting queries to be limited.

1.10 Hints

Hint (1.1)**.** The observed pass rate is 0.8; the half-width is 1.96 standard errors. For (b), solve $\sqrt{0.25/N} = 0.01$.

Hint (1.2)**.** Misses cost $(1-s)$ times the number broken times five days; false alarms cost f times the number good times half a day.

Hint (1.3)**.** The variance left is $1-\rho^2$; the adjustment subtracts θ times the user's excess over last month's average.

Hint (1.4)**.** A fall of two points is a fall of eight questions; the count passed is binomial with 400 trials.

Hint (1.5)**.** Linearity of expectation.

Hint (1.6)**.** Write the risk per checkpoint as $qC_m(1-s(f)) + (1-q)C_f f$ and differentiate in f.

Hint (1.7)**.** Compute t for each candidate. What does the gate's data depend on?

Hint (1.8)**.** $\max(a,b) = \frac{a+b}{2} + \frac{|a-b|}{2}$, and $Z_1 - Z_2$ is normal with variance 2. For the growth, compare with the probability that the maximum exceeds any fixed level.

Hint (1.9)**.** Use Proposition 1.9 with Δ equal to the true fall.

Hint (1.10)**.** Let the loss not depend on the state.

Hint (1.11)**.** Compute $\mathrm{Cov}(D_t, D_{t+1})$ from the shared score S_t. Negative correlation between neighbouring looks makes runs of alarms rarer.

Hint (1.12)**.** When the two means agree, the answer is a function of the training set alone. Each disagreement spends a little of the holdout's privacy; count the disagreements.

1.11 Solutions

Solution (1.1)**.** (a) $\sqrt{0.8 \times 0.2/50} = 0.0566$, a standard error of 5.66 points; the half-width is $1.96 \times 5.66 = 11.09$ points. (b) $N = 0.25/0.01^2 = 2{,}500$ (lab 1c).

Solution (1.2)**.** Ten broken: misses $10 \times 0.4 \times 5 = 20$ days, false alarms $90 \times \frac{1}{90} \times 0.5 = 0.5$ days; with 20.5 against 50 without, worth 29.5 days. One broken: misses 2 days, false alarms $99 \times \frac{1}{90} \times 0.5 = 0.55$ days; with 2.55 against 5, worth 2.45 days. When breakage is common the original evaluation is worth more (42.5 against 29.5 days); when it is rare the quieter one is (2.45 against 1.75). The best threshold depends on the base rate, as Proposition 1.4 says it must (lab 1c).

Solution (1.3)**.** $1-0.6^2 = 0.64$ of the variance remains, as good as $1/0.64 = 1.56$ times the users. The user's adjusted count is $45 - 0.8 \times (40 - 20) = 29$ (lab 1c).

Solution (1.4)**.** The standard deviation of the difference of two scorings is $\sqrt{2 \times 0.25/400} = 0.0354$, that is 3.54 points. A fall of two points or more is a fall of at least eight questions; the exact probability, from the convolution of two binomial distributions, is 0.2980; the normal approximation gives 0.2858 without a continuity correction and 0.2979 with one (threshold 7.5/400). Twenty looks bring $20 \times 0.2980 = 5.96$ false alarms on average (lab 1c).

Solution (1.5)**.** $150 \times 8 \times 0.01 = 12$ alarms on average. No: expectation is linear whatever the dependence. Correlation changes the distribution of the count — alarms arrive in clumps — not its mean.

Solution (1.6)**.** Per checkpoint the risk at operating point f is $R(f) = qC_m(1 - s(f)) + (1-q)C_f f$, so $R'(f) = -qC_m s'(f) + (1-q)C_f$. Concavity makes s' decreasing, so R is convex and is minimised where $R'(f) = 0$, that is $s'(f) = (1-q)C_f/(qC_m)$. Lowering the threshold raises both s and f; it pays while $qC_m\,\Delta s > (1-q)C_f\,\Delta f$. In counts per hundred checkpoints, the extra catches are $100q\,\Delta s$ and the extra false alarms $100(1-q)\,\Delta f$, and the condition reads: extra false alarms per extra catch fewer than C_m/C_f. In Exhibit 1.2, $C_m/C_f = 5/0.5 = 10$.

Solution (1.7)**.** (a) For A, $t = 0.5 + 0.3 \times 0.1 = 0.53$; for B, $t = 0.5 - 0.3 \times 0.05 + 0.2 \times 0.225 = 0.5 - 0.015 + 0.045 = 0.53$. Each user's thumbs-up is a Bernoulli(t) draw, so the gate's data — the counts of thumbs-up in each arm — have the same distribution under A as under B. The standard error of the difference of two rates near one half with 200,000 users per arm is $\sqrt{2 \times 0.25/200{,}000} = 0.00158$; the true difference 0.03 is about nineteen standard errors, so both candidates ship with probability indistinguishable from one. Shipping B lowers quality by 0.05 on the common scale. (b) Any rule based on the gate's data is a function of a random variable whose distribution is identical under A and B; its probability of shipping is therefore identical. No threshold, test or sequential procedure on that data can prefer A. The gate is not merely noisy here: it is structurally blind, because it measures t, and t is the optimiser's target. (c) A measurement of quality that does not pass through users' approval — for instance, expert grading of correctness against references, or a held-out task suite scored by checks of correctness — distinguishes $\Delta q = 0.1$ from $\Delta q = -0.05$. It must not share the thumbs-up signal, nor have been used as a training reward for these candidates (Proposition 1.6).

Solution (1.8)**.** $\mathbb{E}[\max(Z_1, Z_2)] = \mathbb{E}\left[\frac{Z_1+Z_2}{2}\right] + \frac{1}{2}\mathbb{E}|Z_1 - Z_2| = 0 + \frac{1}{2}\sqrt{2}\sqrt{2/\pi} = 1/\sqrt{\pi} = 0.5642$, using $\mathbb{E}|W| = \sigma\sqrt{2/\pi}$ for W normal with mean zero and standard deviation $\sigma = \sqrt{2}$. For growth: for any level c, $P(\max_i Z_i \le c) = \Phi(c)^m \to 0$ as $m \to \infty$, so the maximum eventually exceeds every level. Splitting on the event $\{\max_i Z_i > c\}$, and using $\max_i Z_i \ge Z_1 \ge -|Z_1|$ on its complement, gives $\mathbb{E}[\max_i Z_i] \ge c\,P(\max_i Z_i > c) - \mathbb{E}|Z_1|$; as m grows the probability tends to one, so the expectation eventually exceeds $c - \sqrt{2/\pi}$, for every c. The values 1.1630, 1.5388 and 2.5076 for five, ten and a hundred candidates are from lab 1c.

Solution (1.9)**.** By Proposition 1.9, a true fall Δ triggers a rule with threshold t with probability $\Phi((\Delta - t)/\sigma_N)$. Fixed rule, $t = 2$: $\Phi(-1/0.63) = 0.056$ for a one-point fall and $\Phi(-0.5/0.63) = 0.214$ for one and a half points. Scaled rule, $t = 1.26$: $\Phi(-0.26/0.63) = 0.340$ and $\Phi(0.24/0.63) = 0.648$ (lab 1c). The sentence: at ten thousand questions the noise is two thirds of a point, so a two-point threshold ignores regressions three times the noise.

Solution (1.10)**.** Let $\Theta = \{\theta_1, \theta_2\}$, $A = \{a_1, a_2\}$, and $L(\theta, a_1) = 0$, $L(\theta, a_2) = 1$ for both states; let $X = \theta$ exactly. Then a_1 is optimal under the prior and under every posterior, so by Proposition 1.3 the worth is zero: the evaluation is perfect and the decision does not depend on what it measures. Generally, if a single action minimises $\rho(x, \cdot)$ for almost every x, the equality case of Proposition 1.3 holds and $V(X) = 0$, whatever the accuracy.

Tier III: by standing policy, hints only.

1.12 The Reading

Gemini Team, Google, "Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities", 2025, §2.3. The section on training infrastructure; the replay alarm is one short numbered item of it, in its makers' words. Read them for the economics of an instrument read during a run; the rest of the report is capability, for another day.

Dixit et al., "Silent Data Corruptions at Scale", Facebook, 2021. The anatomy of one silent fault, traced to a single core with clean logs. Skim for the method; savour §§4–5.

Anthropic, "Challenges in evaluating AI systems", 4 October 2023. The bias score of zero, the cost of building and implementing one benchmark, and MMLU's sensitivity to formatting. Short; read it whole.

Anthropic, "A postmortem of three recent issues", 17 September 2025. Serving, not training, but the section "Why detection was difficult" is the best short account in print of evaluations too noisy to see a real fault.

Anthropic, "Introducing Claude 4" and "Activating AI Safety Level 3 protections", both 22 May 2025. The first for the measured reduction in "shortcuts or loopholes", the second for a decision taken because an evaluation could not rule a risk out.

Groeneveld et al., "OLMo: Accelerating the Science of Language Models", 2024, §2.4. An evaluation run in-loop to make decisions of model design, described under the heading "In-Loop Training Ablations".

OLMo Team, "2 OLMo 2 Furious", 2025, §2.5 and §4. The development and held-out discipline, the rationed GSM8K, and the advocacy of declaring the split.

Lambert et al., "Tülu 3: Pushing Frontiers in Open Language Model Post-Training", 2024, §2.2 and §7. The unseen suite, kept unexamined, and the measured overfitting to the development suite.

Llama Team, Meta, "The Llama 3 Herd of Models", 2024, §5.1.1. The paragraph "Significance estimates": the honest error bar, and its honest caveat.

Zheng et al., "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena", NeurIPS 2023 (Datasets and Benchmarks), §3.3. Verbosity bias, defined and measured.

Dwork, Feldman, Hardt, Pitassi, Reingold and Roth, "Generalization in Adaptive Data Analysis and Holdout Reuse", 2015. The danger of reusing a holdout, demonstrated, and Thresholdout; the theory is in "Preserving Statistical Validity in Adaptive Data Analysis" (STOC 2015).

Fabijan et al., "Three Key Checklists and Remedies for Trustworthy Analysis of Online Controlled Experiments at Scale", ICSE-SEIP 2019. The A/A test as a check on the instrument, with its symptoms and remedies.

Deng, Hagar, Stevens, Xifara, Yuan and Gandhi, "From Augmentation to Decomposition: A New Look at CUPED in 2023", 2023. CUPED explained by its first author a decade on. The original — Deng, Xu, Kohavi and Walker, "Improving the Sensitivity of Online Controlled Experiments by Utilizing Pre-Experiment Data", WSDM 2013 — could not be reached for this chapter and is cited as a record only.

Anthropic, "A statistical approach to model evaluations", 19 November 2024. Five recommendations, of which the fourth, paired differences, is this chapter's; the rest are Chapter 3's.